MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: P.L Roll No.: 21 Exp. No.: 1
Name of the Experiment:
Performed on:
Submitted on:

Marks	Teacher's Signature with date
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unit - 01 :- Introduction to Python Programming.

Q1] List the key features and advantages of Python. Also mention any two Python IDEs or editors used for program development.

→ Key features of Python are:-
1. Simple and easy to learn
2. Interpreted language
3. High level language
4. object - oriented
5. Portable

+ Advantages of Python
1. Rapid development
2. open source and free
3. large community support

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

o Demo Python IDEs / Editors.

1 Pycharm 2. IDLE

Q.2] Explain the history and evolution of Python. Describe why Python has become popular. Industry applications.

→ Python was developed by Guido Rossum in the 1980's and released in 1991 It was designed as a high-level interpreted language emphasizing code readability and simplicity.

o Reasons for popularity of Python for Beginners

i] Simple and readable syntax
ii] Easy to learn and understand
iii] Interpreted language with easy debugging

o Reasons for popularity of Python in Industry Applications

i] Versatile language used in multiple domains
ii] Rich set of libraries and frame work.
iii] Plattform independent

Q3] Write a Python Program to accept two numeric Values from the user and perform the following

1] Addition
2] Substraction
3] multiplication
4] Division.

Display the results using proper input and output. prompting.

a = float (input ("enter the first number:"))
b = float (input ("enter the second number:"))

addition = a+b
subtraction = a-b
multiplication = a*b
division = a/b

Displaying results
Print f ("Addition =", addition)
Print f ("Subtraction =", substraction)
Print f ("Multiplication =" Multiplication)
Print f ("Division =" division)

Q4] Analyze the following Python code and Identify the data types of each Variable. Also explain the output.

→ given Code :

```
a=10
b= 5.5
c= 3+4j
d= True
Print (a+b , type (c), d)
```

Data types of Variables

a → int (integer)

b → float (floating - Point number)

c → Complex (complex number with real and imaginary Parts).

d → bool (Boolean value)

output

a+b → 10 + 5.5 = 15.5

type (c) → Displays the data type of Variable

which is < class ' complex ')

d → Prints the Boolean value true.

15.5 < class 'complex' > true.

Q5| Evaluate the use of relational, logical, and bitws operators in python. Justify with suitable example where each type of operator is most appropriate.

Relational operator: used for Comparison used in Conditional and decision Making

a = 10
b = 5
Print (a > b)

o logical operators.
used to combine Conditions.
used in authentication and Validation

x = True
y = false
Print (x & y)

o Bitwise operators
operate on binary numbers
used in low prog and networking

a = 5
b = 3
Print (a & b)

6] Design and implement a menu driven python Program that.

- accepts user input
- performs arthematic, relational, logical and assignment operators.
- Display result clearly

The program should handle int float and boolean data type.

```
a= int (Input ("enter first number:"))
b= int (Input ("enter second number:"))

Print ("1 Arithematic")
Print ("2 Relational")
Print ("3 logical")
Print ("4 Assignment")

ch = int (Input (" Enter Choice:"))
if ch == 1;
    Print ("Add =", a+b)
if
    ch= 2
    Print ("a>b =", a>b)
else if
    ch= 4
    a + = b
    Print ("a =", a)
else
    print ("invalid choice")
```

7 Write a Program in Python to display "Hellow World"

Start

Print "Hellow World"

end

Code

```
Print ("Hellow World")
```

Output

Hello World

Algorithm

1. Start
2. Print Hellow World
3. Stop.

8. Write a Program to Perform average of 5 numbers using input () function and type conversion Concept.

```
a = float(Input ("enter number 1:"))
b = float(Input ("enter number 2:"))
c = float(Input ("enter number 3:"))
d = float(Input ("enter number 4:"))
e = float(Input ("enter number 5:"))

Avg = (a+b+c+d+e)/5
    Print ("average =", avg)
```

Output

```
enter number1 = 10
enter number 2 = 20
enter number 3 = 30
enter number 4 = 40
enter number 5 = 50

Averag = 30.0
```

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Algorithm

1 Start
2. Accept 5 numbers from user
3. Calculate average
4. Display average
5. Stop

Start

Input a, b, c, d, e

Avg = (a+b+c+d+e) / 5

Print Avg

end

Q9 Write a program that prompts user to enter their age and prints out their age in years months and days.

```
age = int (input ("enter age in years:"))
Months = age * 12
days = age * 365
Print ("Age in months : ", months)
Print (" Age in days : ", days)
```

Output

Enter Age in years :18
Age in Months : 216
Age in days = 6570

Start

Input Age

Month = age*12
days = age* 365

Print Months,
days

End.

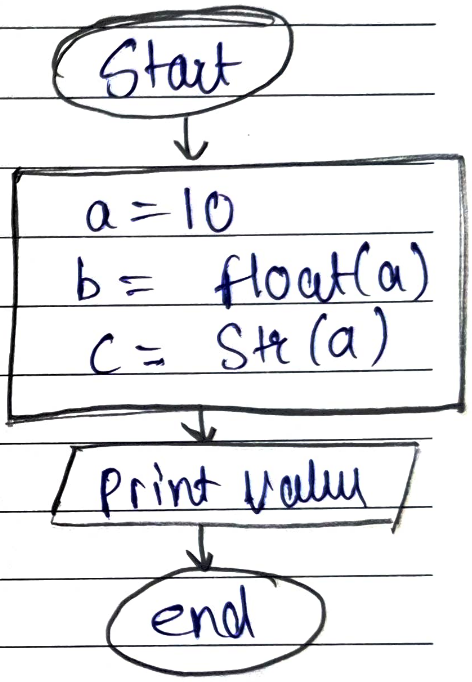
Algorithm

1 Start
2 Input age in Years
3 Calculate Months = age x12
4 Calculate days = age x 365
5 Display Month and days
6 Stop()

Q10 Write a Program to demonstrate the Concept of data types & Conversion.

```
a=10
b= float (a)
c = str (a)

print (a, type (a))
print (b, type (b))
print (c, type (c))
```



Output

```
10 <class 'int'>
10.0 <class 'float'>
10 <class 'str'>
```

Algorithm.

1 Start
2 Assign integer Value
3 Convert integer to float and string
4 Display Values and their data types
5 Stop